

AFYAPLUS HEALTH

AfyaPlus Triage Engine

Week 1 Prototype — From Patient
Messages to Safe Routing Decisions

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Why AfyaPlus Needs This

- Patients write to us in everyday, casual language
- Our systems need clear, consistent data to route each patient correctly

For example: “headache, swelling, blurry vision” from a pregnant patient can signal preeclampsia — we cannot miss it

Left on its own, the AI fails in three ways ↓

THE BUSINESS RISK

**Send a patient to the wrong place,
and you risk their safety — and
system downtime**

1

Chatty, not usable

The AI replies like a chatbot instead of giving data our systems can use.

2

Made-up symptoms

The AI adds symptoms or diagnoses the patient never mentioned.

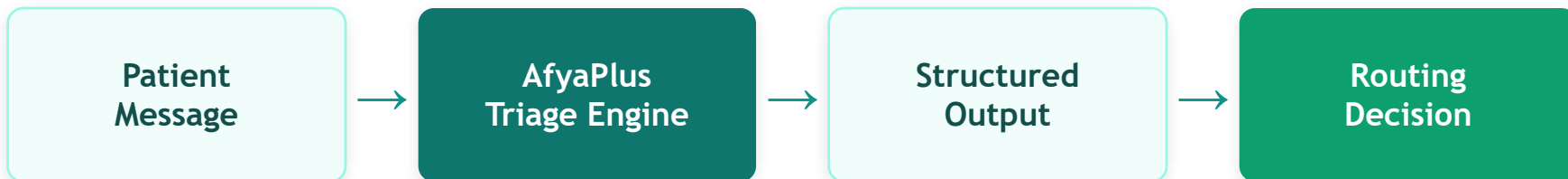
3

Fails when offline

When the internet drops, the whole system freezes instead of carrying on.

What We Built

One patient message in → one structured decision out, every single time.



Every response, same four fields

- ✓ Critical emergency? Yes or No
- ✓ Detected symptoms — only what the patient said
- ✓ One-line reasoning summary
- ✓ Routing destination

Five fixed routing options

Emergency Department

Urgent Care

General Practitioner

Telemedicine

Home Monitoring

Cloud-first with automatic local backup when the internet fails.

Why GPT-4o-mini + llama3.2?

Cloud for quality and speed. Local for resilience and offline safety.

PRIMARY GPT-4o-mini (Cloud)

- Lightweight cloud model — fast and cost-effective
- Strong at following strict rules and complex reasoning
- Average response time ~3.1 seconds in testing

Trade-offs: needs internet, cost per message, data leaves our systems

BACKUP llama3.2 via Ollama (Local)

- Open-source model on our own hardware
- Automatic fallback when the cloud fails or times out
- Works offline — no internet, no per-message cost

Trade-offs: slower (~6.9s), less accurate on complex edge cases

REAL TEST RUNS

Test Case	Path	Response Time	Routing Result
Broken leg, exposed bone	Cloud	2.60 s	Emergency Department
Cold and running nose	Cloud	3.67 s	Home Monitoring
Sore throat (cloud failed)	Local fallback	7.08 s	General Practitioner
Difficulty breathing (cloud failed)	Local fallback	6.69 s	Urgent Care Clinic

How We Control AI Behaviour

Three prompt versions tested — the final one uses role assignment, step-by-step reasoning, and strict guardrails.

The AI is defined as a routing engine

Not a chatbot. Not a doctor.

- ✓ It must follow four fixed reasoning steps before answering
- ✓ Routing is locked to five fixed destinations — it cannot invent new ones
- ✓ Output always follows one fixed structure the backend can read

Guardrails block the AI from:

- ✗ Diagnosing diseases
- ✗ Inventing symptoms the patient didn't mention
- ✗ Giving treatment or medication advice
- ✗ Adding greetings or conversational text

What This Prototype Does NOT Do

Not medically approved

It's a Week 1 prototype — not certified medical software.

No doctor checks it yet

Urgent cases still need a human to review — nothing catches a wrong decision today.

Cloud use shares patient data

When online, messages go to an outside AI service — a privacy issue to fix before launch.

Not for emergencies alone

It should never dispatch an emergency without a person overseeing it.

HONEST ASSESSMENT

This proves the system works. Before real use it needs medical sign-off, a record of every decision, and a required human check for every emergency.

What We Proved — And What's Next

WHAT WE PROVED

- ✓ Working end-to-end triage pipeline
- ✓ Cloud + local fallback resilience demonstrated
- ✓ Structured output produced every time
- ✓ Guardrails reduced fluff and hallucinations
- ✓ Latency baseline documented (cloud ~3s, local ~7s)

NEXT STEPS

- Clinical validation with medical advisors
- Human-in-the-loop review for critical cases
- Audit logging for every triage decision
- Kenya Data Protection Act compliance review
- Integration with AfyaPlus helpline (+254-XXX-XXXX)

The engine works. Now we make it safe for patients.